

220B Homework 2

Solutions

Problem 1: Posterior functionals and the Bayes rule (25 pts)

Let π be a proper prior and let $\pi(\cdot | x)$ denote the posterior given $X = x$, assumed to have a finite first absolute moment for every x .

- (a) Prove that under squared-error loss $L(\theta, a) = (\theta - a)^2$, the Bayes action given x is the posterior mean $\mathbb{E}[\theta | X = x]$, provided the posterior has finite second moment.
- (b) Prove that under absolute-error loss $L(\theta, a) = |\theta - a|$, every posterior median is a Bayes action given x . Give an example (with a discrete posterior) showing the Bayes action need not be unique.
- (c) For $c_1, c_2 > 0$, consider the asymmetric linear loss

$$L(\theta, a) = c_1(\theta - a)\mathbf{1}(a \leq \theta) + c_2(a - \theta)\mathbf{1}(a > \theta).$$

Characterize the Bayes action given x as a posterior quantile, identifying the quantile level in terms of c_1 and c_2 .

Hint (b): split the integral $\int |\theta - a| \pi(d\theta | x)$ at a and differentiate in a using Leibniz; use caution where the posterior has atoms.

Problem 1(a)

The posterior expected loss given x is

$$\rho(a) = \mathbb{E}[(\theta - a)^2 \mid X = x] = \mathbb{E}[\theta^2 \mid X = x] - 2a \mathbb{E}[\theta \mid X = x] + a^2,$$

a strictly convex quadratic in a , finite under the second-moment assumption, minimized at $a^* = \mathbb{E}[\theta \mid X = x]$ with minimum value $\text{Var}(\theta \mid X = x)$. The Bayes action is the posterior mean, and is unique.

Problem 1(b)

Let $F(\cdot | x)$ be the posterior CDF. Because $|\theta - a|$ is convex in a for every θ , the function

$$\rho(a) = \int_{-\infty}^a (a - \theta) \pi(d\theta | x) + \int_a^{\infty} (\theta - a) \pi(d\theta | x)$$

is convex. Its right- and left-derivatives are

$$\rho'_+(a) = 2F(a | x) - 1, \quad \rho'_-(a) = 2F(a^- | x) - 1.$$

The minimum is attained at a iff $\rho'_-(a) \leq 0 \leq \rho'_+(a)$, equivalently $F(a^- | x) \leq \frac{1}{2} \leq F(a | x)$, the definition of a posterior median.

Non-uniqueness. Take posterior $\pi(\theta = 0 | x) = \pi(\theta = 1 | x) = \frac{1}{2}$. Every $a \in [0, 1]$ is a median; direct check: $\rho(a) = \frac{1}{2}a + \frac{1}{2}(1 - a) = \frac{1}{2}$ on $[0, 1]$.

Problem 1(c)

The posterior expected loss is

$$\rho(a) = c_1 \int_a^\infty (\theta - a) \pi(d\theta | x) + c_2 \int_{-\infty}^a (a - \theta) \pi(d\theta | x).$$

By Leibniz,

$$\rho'_+(a) = -c_1(1 - F(a | x)) + c_2 F(a | x) = (c_1 + c_2)F(a | x) - c_1.$$

Convexity makes the first-order condition sufficient, so the minimum is at a with $F(a | x) = c_1/(c_1 + c_2)$. The Bayes action is the posterior $c_1/(c_1 + c_2)$ -quantile; at $c_1 = c_2$ this reduces to the median.

Problem 2: Admissibility of $\delta_0(X) = X$ via Blyth's method (25 pts)

Let $X \sim N(\theta, 1)$ with $\theta \in \mathbb{R}$ and squared-error loss $L(\theta, a) = (\theta - a)^2$. Let $\delta_0(X) = X$.

- (a) Compute $R(\theta, \delta_0)$.
- (b) For each $k > 0$, let π_k be the $N(0, k)$ prior on θ . Derive the Bayes estimator δ^{π_k} and the posterior variance, and show that

$$r(\pi_k, \delta_0) - r(\pi_k, \delta^{\pi_k}) = \frac{1}{k+1}.$$

- (c) Prove that δ_0 is admissible. You may assume that for every rule δ , the function $\theta \mapsto R(\theta, \delta)$ is continuous in θ .

Hint: suppose toward contradiction that δ' dominates δ_0 with $R(\theta^*, \delta') < 1$ at some θ^* . Using the π_k in (b), lower-bound $r(\pi_k, \delta_0) - r(\pi_k, \delta')$ via continuity of $R(\cdot, \delta')$ near θ^* together with the $N(0, k)$ density, and upper-bound it via the Bayes property of δ^{π_k} . Identify the rates of the two bounds in k and send $k \rightarrow \infty$. This is Blyth's (1951) method.

Problem 2(a)

$$R(\theta, \delta_0) = \mathbb{E}_{P_\theta}[(X - \theta)^2] = \text{Var}_{P_\theta}(X) = 1.$$

Problem 2(b)

With $X \mid \theta \sim N(\theta, 1)$ and $\theta \sim N(0, k)$, the joint density factors as

$$\pi(\theta \mid x) \propto \exp\left(-\frac{1}{2}[\theta^2(1/k + 1) - 2\theta x]\right) = \exp\left(-\frac{k+1}{2k}\left[\theta - \frac{k}{k+1}x\right]^2\right),$$

so $\theta \mid X = x \sim N(\frac{k}{k+1}x, \frac{k}{k+1})$. The Bayes estimator is the posterior mean $\delta^{\pi_k}(x) = \frac{k}{k+1}x$; the posterior variance is the constant $\frac{k}{k+1}$. Hence

$$r(\pi_k, \delta^{\pi_k}) = \mathbb{E}[\text{Var}(\theta \mid X)] = \frac{k}{k+1}, \quad r(\pi_k, \delta_0) = \int 1 \pi_k(d\theta) = 1,$$

so $r(\pi_k, \delta_0) - r(\pi_k, \delta^{\pi_k}) = 1 - \frac{k}{k+1} = \frac{1}{k+1}$.

Problem 2(c): lower bound

Suppose toward contradiction that some δ' has $R(\theta, \delta') \leq 1$ for all θ and $R(\theta^*, \delta') < 1$ at some θ^* .

Lower bound. Set $\varepsilon = (1 - R(\theta^*, \delta'))/2 > 0$. By continuity of $\theta \mapsto R(\theta, \delta')$ there exists $\eta > 0$ with $R(\theta, \delta') \leq 1 - \varepsilon$ on $[\theta^* - \eta, \theta^* + \eta]$. Since $R(\theta, \delta_0) - R(\theta, \delta') \geq 0$ everywhere,

$$r(\pi_k, \delta_0) - r(\pi_k, \delta') \geq \varepsilon \int_{\theta^* - \eta}^{\theta^* + \eta} \frac{1}{\sqrt{2\pi k}} e^{-\theta^2/(2k)} d\theta.$$

Let $M = |\theta^*| + \eta$; for $k \geq M^2$, $e^{-\theta^2/(2k)} \geq e^{-1/2}$ on the interval, so for some constant $C > 0$,

$$r(\pi_k, \delta_0) - r(\pi_k, \delta') \geq \frac{C}{\sqrt{k}} \quad (k \geq M^2).$$

Problem 2(c): upper bound and contradiction

Upper bound. δ^{π_k} minimizes Bayes risk under π_k , so $r(\pi_k, \delta') \geq r(\pi_k, \delta^{\pi_k})$, hence

$$r(\pi_k, \delta_0) - r(\pi_k, \delta') \leq r(\pi_k, \delta_0) - r(\pi_k, \delta^{\pi_k}) = \frac{1}{k+1}.$$

Contradiction. For all $k \geq M^2$, $C/\sqrt{k} \leq 1/(k+1)$, equivalently $C(k+1)/\sqrt{k} \leq 1$. As $k \rightarrow \infty$, $(k+1)/\sqrt{k} \rightarrow \infty$, so the left side eventually exceeds 1. No such δ' exists; δ_0 is admissible.

The mechanism is the rate gap: lower bound at $k^{-1/2}$ (Gaussian density at fixed compact set), upper bound at k^{-1} (Bayes-risk gap). The same argument fails in dimension ≥ 3 , where the density rate is $k^{-p/2}$ matching the Bayes-risk gap, which is exactly why Stein shrinkage exists.

Problem 3: Neyman–Scott and the inconsistency of joint Jeffreys (25 pts)

Let $X_{ij} \stackrel{\text{iid}}{\sim} N(\mu_i, \sigma^2)$ for $i = 1, \dots, n$ and $j = 1, 2$, with parameter $(\mu_1, \dots, \mu_n, \sigma^2)$.

Write $T = \frac{1}{2} \sum_{i=1}^n (X_{i1} - X_{i2})^2$.

- (a) Derive $\hat{\sigma}_{\text{MLE}}^2$ in closed form and show $\hat{\sigma}_{\text{MLE}}^2 \xrightarrow{p} \sigma^2/2$. The MLE is inconsistent.
- (b) Compute the Fisher information matrix in the parameterization $(\mu_1, \dots, \mu_n, \sigma)$ and show that $\pi^J(\mu_1, \dots, \mu_n, \sigma) \propto \sigma^{-(n+1)}$.
- (c) Under π^J , show that $\sigma^2 \mid X \sim \text{InvGamma}(n, T/2)$, and that $\mathbb{E}[\sigma^2 \mid X] \xrightarrow{p} \sigma^2/2$ as $n \rightarrow \infty$. The Jeffreys posterior inherits the MLE's inconsistency.
- (d) Under $\pi^{IJ}(\mu_1, \dots, \mu_n, \sigma) \propto 1/\sigma$, derive $\mathbb{E}[\sigma^2 \mid X]$ in closed form and show it is consistent. Comment on treating “joint Jeffreys” as a default when the number of nuisance parameters grows.

Hint: $\sum_{j=1}^2 (X_{ij} - \bar{X}_i)^2 = (X_{i1} - X_{i2})^2/2$ where $\bar{X}_i = (X_{i1} + X_{i2})/2$. In (c), integrate out the μ_i first; the substitution $v = \sigma^2$ then puts the remaining σ -marginal in inverse-gamma form.

Problem 3: setup

Throughout: $\bar{X}_i = (X_{i1} + X_{i2})/2$, $T = \frac{1}{2} \sum_{i=1}^n (X_{i1} - X_{i2})^2$, and $\sum_{j=1}^2 (X_{ij} - \mu_i)^2 = \frac{1}{2} (X_{i1} - X_{i2})^2 + 2(\bar{X}_i - \mu_i)^2$.

Problem 3(a)

Maximizing in μ_i gives $\hat{\mu}_i = \bar{X}_i$. Plugging back,

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{2n} \sum_{i=1}^n \sum_{j=1}^2 (X_{ij} - \bar{X}_i)^2 = \frac{1}{2n} \sum_{i=1}^n \frac{(X_{i1} - X_{i2})^2}{2} = \frac{T}{2n}.$$

Since $X_{i1} - X_{i2} \sim N(0, 2\sigma^2)$, $\mathbb{E}[(X_{i1} - X_{i2})^2] = 2\sigma^2$ and $\mathbb{E}[T] = n\sigma^2$, so $\mathbb{E}[\hat{\sigma}_{\text{MLE}}^2] = \sigma^2/2$. By LLN, $T/n \xrightarrow{p} \sigma^2$, hence $\hat{\sigma}_{\text{MLE}}^2 \xrightarrow{p} \sigma^2/2$. Inconsistent.

Problem 3(b)

From $\log f(x_{ij} \mid \mu_i, \sigma) = -\frac{1}{2} \log(2\pi) - \log \sigma - (x_{ij} - \mu_i)^2 / (2\sigma^2)$, expected second derivatives give per single observation $I_{\mu_i \mu_i}^{(1)} = 1/\sigma^2$, $I_{\sigma \sigma}^{(1)} = 2/\sigma^2$, all cross-terms zero. Aggregating over the two observations per i ,

$$I(\mu_1, \dots, \mu_n, \sigma) = \text{diag}\left(\frac{2}{\sigma^2}, \dots, \frac{2}{\sigma^2}, \frac{4n}{\sigma^2}\right),$$

and $|I| = (2/\sigma^2)^n \cdot (4n/\sigma^2) \propto \sigma^{-2(n+1)}$. Hence $\pi^J \propto |I|^{1/2} \propto \sigma^{-(n+1)}$.

Problem 3(c)

The joint posterior under π^J is

$$\begin{aligned} p(\mu_1, \dots, \mu_n, \sigma \mid X) &\propto \sigma^{-2n} \exp\left(-\frac{1}{2\sigma^2} \sum_{i,j} (X_{ij} - \mu_i)^2\right) \cdot \sigma^{-(n+1)} \\ &= \sigma^{-(3n+1)} \exp\left(-\frac{T+2 \sum_i (\bar{X}_i - \mu_i)^2}{2\sigma^2}\right). \end{aligned}$$

Each μ_i -integral is Gaussian with variance $\sigma^2/2$, contributing $\sigma\sqrt{\pi}$, for a total factor $\propto \sigma^n$. The marginal in σ is

$$p(\sigma \mid X) \propto \sigma^n \cdot \sigma^{-(3n+1)} e^{-T/(2\sigma^2)} = \sigma^{-(2n+1)} e^{-T/(2\sigma^2)}.$$

Substituting $v = \sigma^2$ gives $p(v \mid X) \propto v^{-(n+1)} e^{-T/(2v)}$, which is $\text{InvGamma}(n, T/2)$.
Mean: $\beta/(\alpha - 1) = T/(2(n - 1))$. Therefore

$$\mathbb{E}[\sigma^2 \mid X] = \frac{T}{2(n-1)} = \frac{T}{2n} \cdot \frac{n}{n-1} \xrightarrow{p} \frac{\sigma^2}{2}.$$

The Jeffreys posterior mean inherits the MLE's inconsistency.

Problem 3(d)

Under $\pi^{IJ} \propto 1/\sigma$ the prior contribution is one power of $1/\sigma$ regardless of n . Repeating the marginalization,

$$p(\sigma \mid X) \propto \sigma^n \cdot \sigma^{-(2n+1)} e^{-T/(2\sigma^2)} = \sigma^{-(n+1)} e^{-T/(2\sigma^2)},$$

so $p(v \mid X) \propto v^{-(n+2)/2} e^{-T/(2v)}$, i.e. $\text{InvGamma}(n/2, T/2)$, with mean

$$\mathbb{E}[\sigma^2 \mid X] = \frac{T}{n-2} \xrightarrow{p} \sigma^2.$$

Consistent.

Moral. Joint Jeffreys deposits σ^{-n} from the nuisance parameters; when n grows with sample size, this prior factor overpowers the data and biases the scale by a factor of two. The independence Jeffreys deposits σ^{-1} regardless of dimension and is consistent. “Noninformative” is a property of the prior together with the likelihood, not of the prior alone, and the joint Jeffreys recipe is not a safe default in problems with growing nuisance dimension.

Problem 4: Generalized Bayes rules and improper posteriors (25 pts)

- (a) Let $X \mid \theta \sim N(\theta, 1)$ with formal prior $\pi(\theta) \propto 1$ on $\mathbb{R}$. Verify that the formal posterior $\pi(\theta \mid x) \propto f(x \mid \theta)\pi(\theta)$ is a proper density for every x . Derive the generalized Bayes estimator of θ under squared-error loss and compute its frequentist risk.
- (b) Let $X \mid \theta \sim \text{Binomial}(n, \theta)$ with formal prior $\pi(\theta) \propto \theta^{-1}(1 - \theta)^{-1}$ on $(0, 1)$. Identify the set $A \subseteq \{0, 1, \dots, n\}$ of data values for which the formal posterior fails to be integrable, and justify by explicit integration.
- (c) For $X \in \{0, 1, \dots, n\} \setminus A$, derive the generalized Bayes estimator of θ under squared-error loss in closed form and compare with the MLE $\hat{\theta} = X/n$.

Hint (b): examine the behavior of $\theta^{X-1}(1 - \theta)^{n-X-1}$ near the endpoints $\theta = 0$ and $\theta = 1$.

Problem 4(a)

The formal posterior $\pi(\theta \mid x) \propto \exp(-(x - \theta)^2/2)$ is the $N(x, 1)$ density in θ and integrates to 1. Proper. Under squared-error loss the generalized Bayes estimator is the posterior mean $\delta^{GB}(x) = x$, with frequentist risk $R(\theta, \delta^{GB}) = \text{Var}_{P_\theta}(X) = 1$.

Problem 4(b)

The formal posterior is $\pi(\theta \mid X) \propto \theta^{X-1}(1-\theta)^{n-X-1}$ on $(0, 1)$. The integral $\int_0^1 \theta^{\alpha-1}(1-\theta)^{\beta-1}d\theta$ is finite iff $\alpha > 0$ and $\beta > 0$. Here $\alpha = X$, $\beta = n - X$. Divergence at $\theta = 0$ when $X = 0$ ($\alpha = 0$); divergence at $\theta = 1$ when $X = n$ ($\beta = 0$). Hence $A = \{0, n\}$.

Problem 4(c)

For $X \in \{1, \dots, n-1\}$ the posterior is $\text{Beta}(X, n-X)$, with mean

$$\delta^{GB}(X) = \frac{X}{X + (n-X)} = \frac{X}{n} = \hat{\theta}_{\text{MLE}}.$$

The Haldane prior recovers the MLE on the interior, and is undefined at the boundary $X \in \{0, n\}$ where the MLE is also degenerate.